University of Information Technology & Sciences

Department of Computer Science & Engineering

BSC in CSE- 3rd Year ,2nd semester (Spring 2026)

CSE0611323: Microprocessors,Microcontrollers and Assembly Language

Section: 6B1

Course Information

Item	Details
Course Code	CSE0611323
Course Title	Microprocessors,Microcontrollers and Assembly Language
Section	6B1
Experiment -02 Title	Arithmetic operation of two one digit values in assembly language
Submission Date	16 August 2026
Course Teacher	Tahmina Yasmin Lima, Lecturer

Student Information

Information	Details
Student Name	Helal Uddin Patwary
Student ID	0432410005101086
Section	6B1

Introduction: In this Assignment-2 ,Performs basic arithmetic operations on two one-digit numbers using 8086 Assembly Language.

- Takes two digits as keyboard input from the user.
- Performs:
  - Addition
  - Subtraction
  - Multiplication
  - Division
  - Remainder
- Uses DOS interrupt 21h for keyboard input and screen output.
- Demonstrates important 8086 instructions:
  - `MOV` — transfers data
  - `ADD` — performs addition
  - `SUB` — performs subtraction
  - `MUL` — performs multiplication
  - `DIV` — performs division
  - `INT` — invokes DOS services
- Converts keyboard input from ASCII to numeric value by subtracting `30h`.
- Converts numeric results back to ASCII by adding `30h` before displaying them.

CODE:

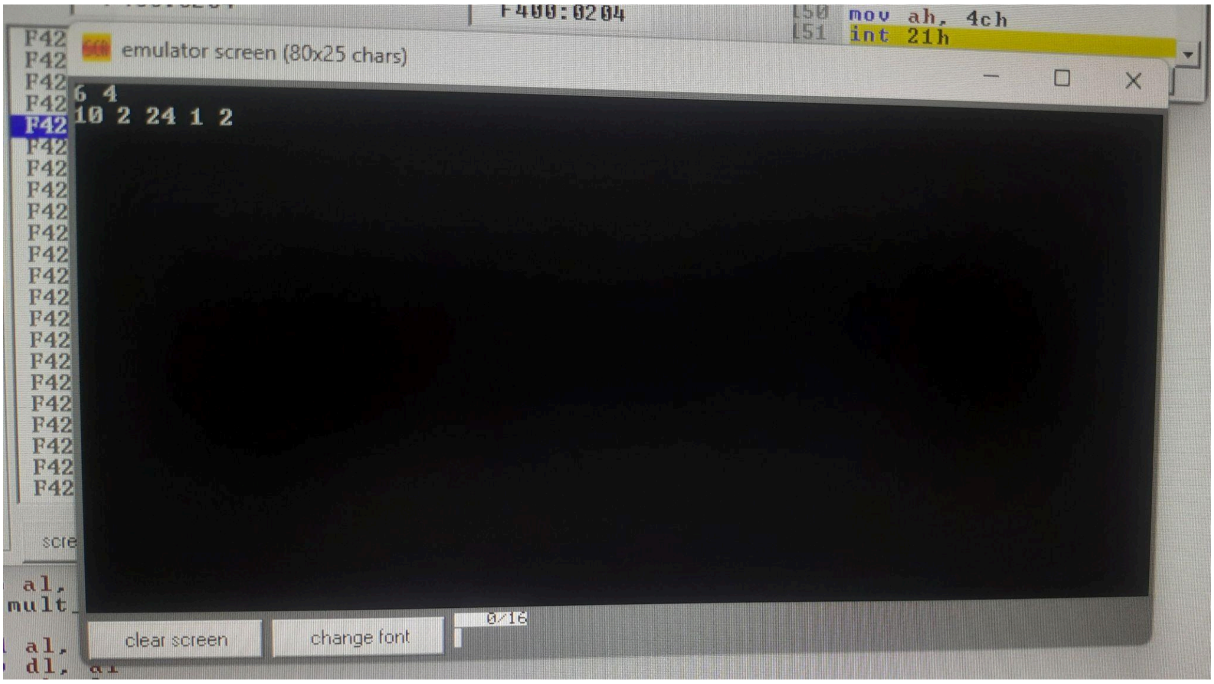
```
.model small
.stack 100h

.data
   n1   db ?
   n2   db ?
   sum  db ?
   dif  db ?
   mult db ?
   quot db ?
   rem  db ?

.code
main proc

   mov ax, @data
   mov ds, ax

   mov ah, 1
   int 21h
   sub al, 30h
   mov n1, al

   mov ah, 2
   mov dl, 32
   int 21h

   mov ah, 1
   int 21h
```

```
    sub al, 30h
    mov n2, al

    mov al, n1
    add al, n2
    mov sum, al

    mov al, n1
    sub al, n2
    mov dif, al

    mov al, n1
    mov bl, n2
    mul bl
    mov mult, al

    mov al, n1
    mov ah, 0
    mov bl, n2
    div bl
    mov quot, al
    mov rem, ah

    mov ah, 2
    mov dl, 13
    int 21h

    mov dl, 10
    int 21h

    mov al, sum
    mov ah, 0
    mov bl, 10
    div bl

    mov cl, ah

    cmp al, 0
    je sum_one

    add al, 30h
    mov dl, al
    mov ah, 2
    int 21h

sum_one:
    mov dl, cl
    add dl, 30h
    mov ah, 2
    int 21h

    mov dl, 32
    int 21h

    mov al, n1
    cmp al, n2
    jae dif_positive
```

```
    mov dl, '-'
    mov ah, 2
    int 21h

    mov al, n2
    sub al, n1
    add al, 30h
    mov dl, al
    mov ah, 2
    int 21h

    jmp dif_done

dif_positive:
    mov dl, dif
    add dl, 30h
    mov ah, 2
    int 21h

dif_done:

    mov dl, 32
    mov ah, 2
    int 21h

    mov al, mult
    mov ah, 0
    mov bl, 10
    div bl

    mov cl, ah

    cmp al, 0
    je mult_one

    add al, 30h
    mov dl, al
    mov ah, 2
    int 21h

mult_one:
    mov dl, cl
    add dl, 30h
    mov ah, 2
    int 21h

    mov dl, 32
    int 21h

    mov dl, quot
    add dl, 30h
    mov ah, 2
    int 21h

    mov dl, 32
    int 21h
```

```
    mov dl, rem
    add dl, 30h
    mov ah, 2
    int 21h

    mov ah, 4ch
    int 21h

main endp
end main
```

Output Screenshot:



I learned:

- How to perform multiplication using `MUL`.
- How to perform division using `DIV`.
- How to get the quotient and remainder from division.
- How to display two-digit results such as `10` and `24`.
- How to convert multi-digit numerical results into ASCII for output.
- How to handle arithmetic results using 8086 registers.

Conclusion

The program successfully performs basic arithmetic operations on two one-digit numbers using 8086 Assembly Language. It demonstrates the use of `ADD`, `SUB`, `MUL`, and `DIV` instructions along with registers and memory variables. The program also shows how keyboard input is converted from ASCII to numeric values using `30h` and how the results are displayed through DOS interrupt `21h`. Overall, it provides a clear practical understanding of basic arithmetic operations and input/output handling in 8086 Assembly Language.